ECA0806- ANALOG AND DIGITAL COMMUNICATION

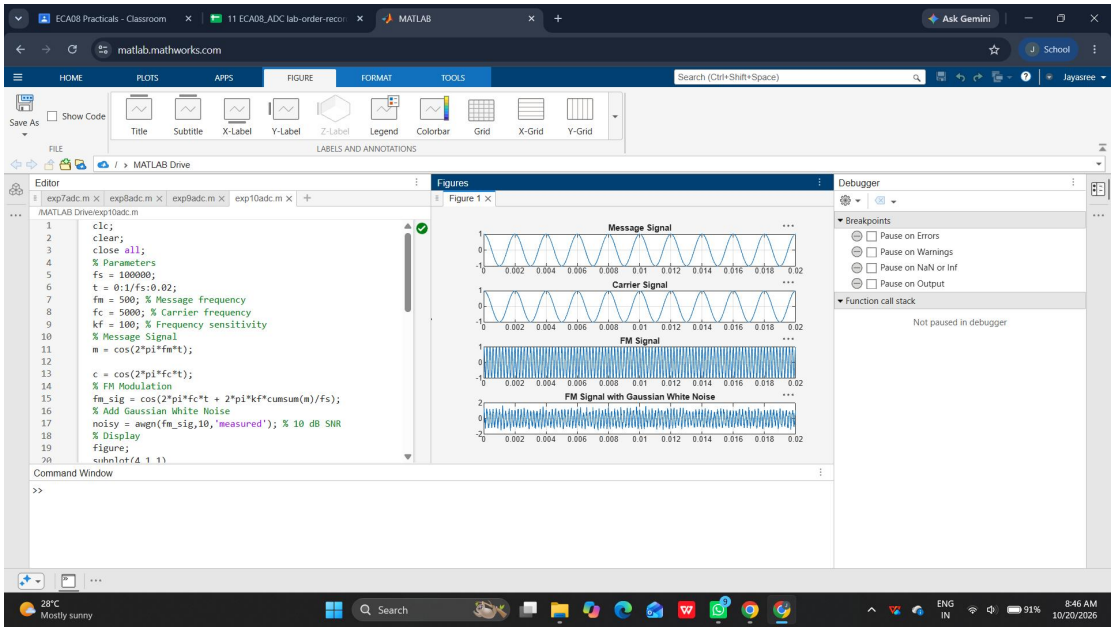
EXPERIMENT - 10

NAME : JAYASREE S (192512299)

PROGRAM:

```
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500; % Message frequency
fc = 5000; % Carrier frequency
kf = 100; % Frequency sensitivity
% Message Signal
m = cos(2*pi*fm*t);

c = cos(2*pi*fc*t);
% FM Modulation
fm_sig = cos(2*pi*fc*t + 2*pi*kf*cumsum(m)/fs);
% Add Gaussian White Noise
noisy = awgn(fm_sig,10,'measured'); % 10 dB SNR
% Display
figure;
subplot(4,1,1)
plot(t,m)
title('Message Signal')
grid on
subplot(4,1,2)
plot(t,m)
title('Carrier Signal')
grid on
subplot(4,1,3)
plot(t,fm_sig)
title('FM Signal')
grid on
subplot(4,1,4)
plot(t,noisy)
title('FM Signal with Gaussian White Noise')
grid on
```


ECA08 Practicals - Classroom
11 ECA08_ADC lab-order-recor
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Y-Grid
FILE
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MATLAB Drive
Editor
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exp8adc.m
exp9adc.m
exp10adc.m
/MATLAB Drive/exp10adc.m
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Figures
Figure 1
Message Signal
Carrier Signal
FM Signal
FM Signal with Gaussian White Noise
Debugger
Breakpoints
Pause on Errors
Pause on Warnings
Pause on NaN or Inf
Pause on Output
Function call stack
Not paused in debugger
Command Window
>>
28°C
Mostly sunny
Search
ENG
IN
91%
8:46 AM
10/20/2026